

RIDA MELKAOUI

INDUSTRIAL ENGINEER | DATA, AI & BUSINESS INTELLIGENCE

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PROFILE

Industrial and Production Engineer combining supplier-quality ownership with data analytics, BI and applied AI. Builds traceable decision systems with Python, SQL, Power BI and statistical/optimization methods - from launch and issue visibility to demand planning, routing and AI-agent reliability.

PROFESSIONAL EXPERIENCE

MAGNA INTERNATIONAL <i>Supplier Quality Engineer - Electronics & Metal Commodities</i>	Kenitra, Morocco Mar 2026 - Present
- Manage supplier quality across Europe, Asia and Morocco; use performance indicators, launch readiness, 8D/CAPA status and open-issue risk to prioritize action. - Developed an SQE BI system covering nomination through Safe Launch and SOP, consolidating activities, critical-part status, milestones and open issues into one traceable management view. - Built a smart notification pipeline for overdue actions and launch risks, keeping owners and deadlines visible across the project portfolio.	
STELLANTIS R&D <i>Engineering Intern - Data Analytics, Process Optimization & AI</i>	Casablanca, Morocco Feb 2025 - Aug 2025
- Mapped a configuration-anomaly workflow and built a Python application with a structured Databricks data model for detection, prioritization and traceability; reduced treatment time from 2 days to 2 minutes.	
SANY - FORGE DE BAZAS <i>Engineering Intern - Process Digitalization & Maintenance</i>	Casablanca, Morocco Jul 2024 - Sep 2024
- Built a Unity/NLP assistant to structure access to maintenance and spare-parts knowledge, reducing fragmented search and improving shop-floor information flow.	
3D SMART FACTORY <i>Data Analyst Intern</i>	Mohammedia, Morocco Aug 2023 - Sep 2023
- Built Power BI and Excel reporting around OEE, FPY, scrap and downtime to expose production bottlenecks for Lean review.	

SELECTED DECISION SYSTEMS

Demand / Order Forecasting and replenishment Built a cutoff-safe Python decision engine across 900 M5 item-store series; improved WAPE 4.5% vs the strongest calibration-selected seasonal baseline and simulated 1,528 fewer lost units at 96.7% fulfilled demand. Evidence
Route / Control Optimization and operations Learned zone sequencing from 6,112 historical routes and built a constrained route policy; on a separate 13-route public cohort, cut simulated drive time 2.8% vs nearest neighbour, maintained 100% timed-package adherence and eliminated 332 zone re-entries. Evidence
Agent / Proof AI evaluation and observability Built a reliability lab over 3,336 official historical tool-agent trajectories; implemented Pass1-Pass4, cost, latency and failure diagnostics, exposing a 14.9-point one-shot-to-repeatability loss in the pinned GPT-4.1 telecom cohort. Evidence

EDUCATION

Mohammed VI International Civil Aviation Academy (AIAC)	Nouaceur, Morocco
Engineering Degree - Industrial & Production Engineering	2022 - 2025
Salmane El Farissi Preparatory Classes	Sale, Morocco
MPSI - MP	2020 - 2022

SKILLS

Data & AI: Python, SQL, Pandas, NumPy, scikit-learn, statistics, forecasting, experimentation, data modeling, data quality, Databricks, Git, Docker.

BI & Delivery: Power BI, DAX, Excel, Power Query, Tableau, React, TypeScript, FastAPI, dashboard design, KPI systems, automation, reporting.

Industrial & Quality: supplier/process performance, PPM, OEE, FPY, scrap, downtime, 8D/CAPA, FMEA, root-cause analysis, Lean, launch/SOP governance.

LANGUAGES & CERTIFICATIONS

Languages: Arabic - Native | French - Fluent | English - C1 (TOEIC 855) Certifications: [IBM Data Analyst Professional Certificate \(2026\)](#); [EF SET C1](#); [TOEIC 855](#)